1. Linear Equations in Linear Algebra

1.1 system of linear equations

A linear equation: (ax_1 + a_2x_2 + $...$ +a_nx_n =b), where coefficients (a_1, a_2, $...$, a_n) are real or complex numbers.

A system of linear equations (linear system): A collection of one or more linear equations involving the same variables.

Equivalent linear systems: Two linear systems are equivalent if they have the same solution set.

A system of linear equations has:

- No solution
- Exactly one solution
- Infinitely many solutions

Given a linear system：

$$\begin{cases} x_1 - 2x_2 + x_3 = 0 \\ 2x_1 - 8x_3 = 8 \\ 5x_1 - 5x_3 = 10 \end{cases}$$

Coefficient matrix（系数矩阵）：

$$\begin{pmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ 5 & 0 & -5 \end{pmatrix}$$

Augmented matrix（增广矩阵）：

$$\begin{pmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{pmatrix}$$

Elementary row operations

1. Replace one row by the sum of itself and a multiple of another row
2. Interchange two rows
3. Multiply all entries in a row by a nonzero constant

Row equivalent matrices: Two matrices are row equivalent if there is a sequence of elementary row operations that transforms one matrix into another.

If the augmented matrices of two linear systems are row equivalent, then they have the same solution set.

1.2 Row Reduction and Echelon Forms

Definition:

A matrix is in echelon form if:

1. All nonzero rows are above any rows of all zeros.
2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
3. All entries in a column below a leading entry are zeros.

If a matrix in echelon form satisfies the following additional conditions, then it is in reduced echelon form (REF):

4. The leading entry in each nonzero row is 1.
5. Each leading 1 is the only nonzero entry in its column.

Theorem 1: Uniqueness of the Reduced Echelon Form

Each matrix is row equivalent to one and only one reduced echelon matrix.

Definition: Pivot

- A pivot position in a matrix (A) is a location in (A) that corresponds to a leading 1 in the reduced echelon form of (A).
- A pivot column is a column of (A) that contains a pivot position.

The variables corresponding to pivot columns in the matrix are basic variables.
The other variables are free variables.

Theorem 2: Existence and Uniqueness Theorem

A linear system is consistent if and only if the rightmost column of the augmented matrix is not a pivot column — that is, if and only if an echelon form of the augmented matrix has no row of the form $[0 \quad \dots \quad 0 \quad b] \quad (\text{with } b \neq 0)$

If a linear system is consistent, then the solution set contains either:

- (1) a unique solution, when there are no free variables, or
- (2) infinitely many solutions, when there is at least one free variable.

1.3 Vector Equations

Parallelogram Rule for addition

If $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^2$ are represented as points in the plane, then $\vec{u}$ + $\vec{v}$ corresponds to the fourth vertex of a parallelogram whose other vertices are $\vec{0}$, $\vec{u}$, and $\vec{v}$.

Algebraic Properties of $\mathbb{R}^n$

For all $\vec{u}$, $\vec{v}$, $\vec{w}$ in $\mathbb{R}^n$, and all scalars c and d:

1. **Commutativity**: $\vec{u} + \vec{v} = \vec{v} + \vec{u}$
2. **Associativity**: $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$
3. **Zero Vector Property**: $\vec{u} + \vec{0} = \vec{0} + \vec{u} = \vec{u}$
4. **Negative Vector Property**: $\vec{u} + (-\vec{u}) = -\vec{u} + \vec{u} = \vec{0}$

- Note: $-\vec{u}$ denotes $(-1)\vec{u}$

5. **Distributivity of Scalar Multiplication over Vector Addition**: $c(\vec{u}+\vec{v}) = c\vec{u} + c\vec{v}$
6. **Distributivity of Scalar Multiplication over Scalar Addition**: $(c+d)\vec{u} = c\vec{u} + d\vec{u}$
7. **Associativity of Scalar Multiplication**: $c(d\vec{u}) = (cd)\vec{u}$
8. **Identity Property of Scalar Multiplication**: $1\vec{u} = \vec{u}$

Linear Combination

Given vectors $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p$ in $\mathbb{R}^n$ and given scalars $c_1, c_2, \ldots, c_p$, the vector $\vec{y}$ defined by:

$$\vec{y} = c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_p\vec{v}_p$$

is called a **linear combination** of $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p$ with weights $c_1, c_2, \ldots, c_p$.

Vector Equation & Linear System Relationship

A vector equation:

$$x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{b}$$

has the same solution set as the linear system whose augmented matrix is:

$$\begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \ldots & \vec{a}_n & \vec{b} \end{bmatrix}$$

In particular:
$\vec{b}$ can be generated by a linear combination of $\vec{a}_1, \ldots, \vec{a}_n$ **if and only if** there exists a solution to the linear system corresponding to the above matrix.

Definition: Span

If $\vec{v}_1, \ldots, \vec{v}_p$ are in $\mathbb{R}^n$, then the set of all linear combinations of $\vec{v}_1, \ldots, \vec{v}_p$ is denoted by $\text{Span}\{\vec{v}_1, \ldots, \vec{v}_p\}$ and is called the subset of $\mathbb{R}^n$ spanned (generated) by $\vec{v}_1, \ldots, \vec{v}_p$.

That is, $\text{Span}\{\vec{v}_1, \ldots, \vec{v}_p\}$ is the collection of all vectors that can be written in the form $c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_p\vec{v}_p$ with $c_1, \ldots, c_p$ scalars.

Asking whether the vector equation $x_1\vec{v}_1 + \cdots + x_p\vec{v}_p = \vec{b}$ has a solution is equivalent to asking whether $\vec{b}$ is in $\text{Span}\{\vec{v}_1, \ldots, \vec{v}_p\}$.

In particular, the vector $\vec{b}$ is in $\text{Span}\{\vec{v}_1, \ldots, \vec{v}_p\}$ if and only if the augmented matrix $\begin{bmatrix} \vec{v}_1 & \ldots & \vec{v}_p & \vec{b} \end{bmatrix}$ has a solution.

Particularly,the zero vector $\vec{0}$ must be in $\text{Span}\{\vec{v}_1, \ldots, \vec{v}_p\}$.

1.4 The Matrix Equation $A\vec{x} = \vec{b}$

Definition:

If A is an $m \times n$ matrix, with columns $\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n$, and if $\vec{x}$ is in $\mathbb{R}^n$, then the product of A and $\vec{x}$, denoted by $A\vec{x}$, is the linear combination of the columns of A using the corresponding entries in $\vec{x}$ as weights, that is:

$$A\vec{x} = x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n$$

Theorem 3:

If A is an $m \times n$ matrix, with columns $\vec{a}_1, \ldots, \vec{a}_n$, and if $\vec{b}$ is in $\mathbb{R}^m$, then the matrix equation $A\vec{x} = \vec{b}$ has the same solution set as the vector equation $x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{b}$, which, in turn, has the same solution set as the system of linear equations whose augmented matrix is $\begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \ldots & \vec{a}_n & \vec{b} \end{bmatrix}$.

The equation $A\vec{x} = \vec{b}$ has a solution if and only if $\vec{b}$ is a linear combination of the columns of A.

Theorem 4:

Let A be an $m \times n$ matrix. Then the following statements are logically equivalent (for a particular A, either all are true or all are false):
a. For each $\vec{b}$ in $\mathbb{R}^m$, $A\vec{x} = \vec{b}$ has a solution.
b. Each $\vec{b}$ in $\mathbb{R}^m$ is a linear combination of the columns of A.
c. The columns of A span $\mathbb{R}^m$.
d. A has a pivot position in every row.

Row-Vector Rule for computing $A\vec{x}$

It refers to the linear combination of the rows of A with weights from the vector $\vec{x}$.

Theorem 5

If A is an $m \times n$ matrix, $\vec{u}$ and $\vec{v}$ are vectors in $\mathbb{R}^n$,and c is a scalar,then:
a. A($\vec{u} + \vec{v}$) = $A\vec{u}$ + $A\vec{v}$
b. A($c\vec{u}$) = c(A$\vec{u}$)

1.5 Solution Sets of Linear Systems

Homogeneous linear equations: $A\vec{x} = \vec{0}$
trivial solution: $\vec{x} = \vec{0}$
nontrivial solution: $\vec{x} \neq \vec{0}$

The homogeneous equation $A\vec{x} = \vec{0}$ has a nontrivial solution if and only if the equation has at least one free variable.

When the solution set is written as a linear combination of vectors, we say that the solution is in parametric vector form.

If the solution set of a homogeneous linear equation $A\vec{x} = \vec{0}$ is written as $\mathrm{Span}\{\vec{u}_1, \ldots, \vec{u}_p\}$, then the solution set can be written as $\vec{x} = c_1\vec{u}_1 + c_2\vec{u}_2 + \cdots + c_p\vec{u}_p$.

Theorem 6:

Suppose the equation $A\vec{x} = \vec{b}$ is consistent for some given $\vec{b}$, and let $\vec{x}_p$ be a particular solution. Then the solution set of $A\vec{x} = \vec{b}$ is the set of all vectors of the form $\vec{x} = \vec{x}_p + \vec{v}_h$, where $\vec{v}_h$ is any solution of the homogeneous equation $A\vec{x} = \vec{0}$.

1.7 Linear Independence

Definition:

An indexed set of vectors $\{\vec{v}_1, \ldots, \vec{v}_p\}$ in $\mathbb{R}^n$ is said to be linearly independent if the vector equation

$$x_1\vec{v}_1 + x_2\vec{v}_2 + \cdots + x_p\vec{v}_p = \vec{0}$$

has only the trivial solution.

The set $\{\vec{v}_1, \ldots, \vec{v}_p\}$ is said to be linearly dependent if there exist scalars $c_1, \ldots, c_p$, not all zero, such that

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_p\vec{v}_p = \vec{0}$$

The columns of a matrix A are linearly independent if and only if the equation $A\vec{x} = \vec{0}$ has only the trivial solution.

A set of two vectors $\{\vec{v}_1, \vec{v}_2\}$ is linearly dependent if at least one of the vectors is a multiple of the other.

The set $\{\vec{v}_1, \vec{v}_2\}$ is linearly independent if and only if neither of the vectors is a multiple of the other.

Theorem 7: Characterization of Linearly Dependent Sets

An indexed set $S = \{\vec{v}_1, \ldots, \vec{v}_p\}$ of two or more vectors is linearly dependent if and only if at least one of the vectors in S is a linear combination of the others.

In fact, if S is linearly dependent and $\vec{v}_1 \neq \vec{0}$, then some $\vec{v}_j$ (with $j \geq 1$) is a linear combination of the preceding vectors $\vec{v}_1, \ldots, \vec{v}_{j-1}$.

Theorem 8:

If a set contains more vectors than there are entries in each vector, then the set is linearly dependent. That is, any set of p vectors in $\mathbb{R}^n$ is linearly dependent if $p > n$.

Theorem 9:

If a set S = $\mathrm{Span}\{\vec{u}_1, \ldots, \vec{u}_p\}$ in $\mathbb{R}^n$ contains the zero vector,then the set is linearly dependent.

1.8 Introduction to Linear Transformation

A transformation (or function/mapping) T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns to each $\vec{x}$ in $\mathbb{R}^n$ a vector $T(\vec{x})$ in $\mathbb{R}^m$.

- The set $\mathbb{R}^n$ is the domain of T, and the set $\mathbb{R}^m$ is the codomain of T.
- For $\vec{x}$ in $\mathbb{R}^n$, the vector $T(\vec{x})$ in $\mathbb{R}^m$ is the image of $\vec{x}$ under the transformation T.
- The set of all images $T(\vec{x})$ is the range of T.

A transformation (or mapping) T is linear if:

1. $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ for all $\vec{u}, \vec{v}$ in the domain of T.
2. $T(c\vec{u}) = cT(\vec{u})$ for all scalars c and all $\vec{u}$ in the domain of T.

If T is a linear transformation, then:

1. $T(0) = 0$

2. $T(c_1\vec{u}_1 + c_2\vec{u}_2) = c_1T(\vec{u}_1) + c_2T(\vec{u}_2)$
3. $T(c_1\vec{u}_1 + \cdots + c_n\vec{u}_n) = c_1T(\vec{u}_1) + \cdots + c_nT(\vec{u}_n)$

1.9 The Matrix of a Linear Transformation

Theorem 10:

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then there exists a unique matrix A such that

$T(\vec{x}) = A\vec{x}$ for all $\vec{x}$ in $\mathbb{R}^n$.

In fact, A is the $m \times n$ matrix whose j-th column is the vector $T(\vec{e}_j)$, where $\vec{e}_j$ is the j-th column of the identity matrix in $\mathbb{R}^n$. That is:

$$A = [T(\vec{e}_1)\ T(\vec{e}_2)\ \ldots\ T(\vec{e}_n)]$$

Definition: Surjective

A mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ is said to be onto if every vector in $\mathbb{R}^m$ is the image of at least one vector in $\mathbb{R}^n$.

1. existence question (at least one solution)
2. columns $\geq$ rows , and rank $=$ row

Definition: Injective

A mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ is said to be one-to-one if each vector in $\mathbb{R}^m$ is the image of at most one vector in $\mathbb{R}^n$.

1. uniqueness question(at most one solution)
2. columns $\leq$ row, rank columns.

Theorem 11:

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then T is one-to-one if and only if the equation $T(\vec{x}) = \vec{0}$ has only the trivial solution.

Theorem 12:

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation and let A be the standard matrix for T. Then:

- T is surjective if and only if the columns of A span $\mathbb{R}^m$.
- T is injective if and only if the columns of A are linearly independent.